MUGE Compression Cycle 2: Unified F_env Modular 7-System Environmental Calculator

Daniel T. Murphy

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Classification: FIRST UQFF multi-system compression cycle 2 framework; FIRST unified F_env modular architecture across 7 canonical astrophysical classes
Author: Daniel T. Murphy
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<!— UQFF constants: κ = 5.0e-4 day-1, [SSq] = 0.57, H_SCm $\approx$ 0.99, U_UA $\approx$ 0.0001 —>

Abstract

MUGE Compression Cycle 2 formalises the multi-system unified gravitational calculator, compressing the per-system equations established across Sessions 1–114 into a single modular architecture. This paper documents the 7-system root registry: MagnetarSGR1745, SagittariusA, TapestryStarbirth, Westerlund2, PillarsCreation, RingsRelativity, and UniverseGuide — each contributing system-specific F_env terms that sum to the total environmental gravitational modifier. The framework introduces the critical distinction between the **compressed MUGE** (analytical one-line forms) and the **full-form UQFF** (explicit component summation), validating that both yield the same g_UQFF to within numerical precision.

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2. Core Architecture — PAPER_452

2.1 Compression Principle

"Compression" means reducing per-system individuated equations into a **modular function call registry**. Instead of 7 separate classes, a single compressed equation calls system-specific F_env modules:

$$g_{\mathrm{UQFF}}^{(j)}(t) = \frac{GM_j(t)}{r_j^2}(1 + H_z t)(1 - B_j/B_{\mathrm{crit}}) + \sum_k U_{gk}^{(j)} + F_{\mathrm{env}}^{(j)}(t)$$

The **compressed form** replaces the explicit Ug sum with a pre-tabulated module value:

$$g_{\mathrm{UQFF}}^{(j),\mathrm{comp}}(t) = g_{
m DPM}^{(j)}(1 + H_z t) + F_{\mathrm{env}}^{(j)}$$

2.2 7-System Registry

#	System	M (kg)	r (m)	B (T)	F_env type
1	MagnetarSGR1745	5.58$\times$1030 (2.8 MM_sun)	1$\times$104	1$\times$1011	B_field saturation
2	SagittariusA	8.17$\times$1036 (4.1$\times$106 MM_sun)	6$\times$109	1$\times$10-3	SMBH accretion disk
3	TapestryStarbirth	9.96$\times$1033 (500 MM_sun)	1$\times$1016	1$\times$10-5	SFR + outflow
4	Westerlund2	1.99$\times$1034 (104 MM_sun)	6$\times$1016	1$\times$10-5	Stellar wind
5	PillarsCreation	3.98$\times$1032 (200 MM_sun)	6$\times$1016	1$\times$10-5	Radiation P_rad
6	RingsRelativity	1$\times$1039	1$\times$1020	1$\times$10-6	Lensing shear
7	UniverseGuide	1$\times$1053	4.4$\times$1026	~0	Full F_cosmo

2.3 F_env Per-System Equations

1. MagnetarSGR1745 F_env:

$$F_{\mathrm{env,Mag}} = U_{A}\rho_{\mathrm{vac}}\left(1 + B/B_{\mathrm{crit}}\right) - U_{g4,\mathrm{mag}}$$

Where $B_{\mathrm{crit}} = 4.4\times10^{13}$ T, $B/B_{\mathrm{crit}} = 10^{11}/4.4\times10^{13} \approx 2.27\times10^{-3}$

2. SagittariusA F_env (SMBH accretion):

$$F_{\mathrm{env,SgrA}} = \frac{GM_{\mathrm{disk}}}{r_{\mathrm{ISCO}}^2}\cdot f_{\mathrm{acc}} + \Omega_{\mathrm{disk}}^2 r_{\mathrm{disk}}$$

Where r_ISCO = 6GMc-2 = innermost stable circular orbit.

3. TapestryStarbirth F_env:

$$F_{\mathrm{env,Tap}} = {\mathrm{SFR}}\cdot{v_{\mathrm{wind}}^2/r} + P_{\mathrm{outflow}}$$

4. Westerlund2 F_env (stellar wind):

$$F_{\mathrm{env,W2}} = \rho_{\mathrm{fluid}} v_{\mathrm{wind}}^2 = 10^{-20}\times(10^4)^2 = 10^{-12}\ \mathrm{m}/s^2$$

5. PillarsCreation F_env (radiation):

$$F_{\mathrm{env,Pill}} = \frac{L_{\mathrm{cluster}}}{4\pi r^2 c}\cdot\frac{\rho}{m_H}$$

6. RingsRelativity F_env (gravitational lensing shear):

$$F_{\mathrm{env,Rings}} = \frac{4GM}{c^2 r}\cdot\frac{d_{\mathrm{S}}d_{\mathrm{LS}}}{d_{\mathrm{L}}} \quad [\text{lensing convergence}]$$

7. UniverseGuide F_env (full cosmic):

$$F_{\mathrm{env,Univ}} = g_{\mathrm{QG}} + g_{\mathrm{DM}} + g_{\mathrm{GW}} = F_{\mathrm{cosmo}}(t)$$

2.4 Total Compressed System Sum

$$G_{\mathrm{total}}(t) = \sum_{j=1}^{7} g_{\mathrm{UQFF}}^{(j),\mathrm{comp}}(t)$$

This is the **state vector** of the 7-system universe at cosmic time t.

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3. Ug3 Compressed Form

The standard Ug3 string-rotation term is replaced in Cycle 2 by the compressed form:

$$U_{g3}' = \frac{GM_{\mathrm{ext}}}{r_{\mathrm{ext}}^2}$$

Where M_ext and r_ext are the **external mass and radius** contributing to cross-system string tension. This simplified form discards the $(1 + v_s/c \cos\theta)$ angular factor for the Cycle 2 compression (angle-averaged over the ensemble).

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4. Psi_total in Compression Cycle 2

$$\psi_{\mathrm{total}}^{(7)} = \int_0^\infty A(k)e^{i(kr-\omega t)} dk + \frac{[SSq]^{n_{26}}}{[SSq]^{n_{26}-1}}$$

The second term is the UQFF **quantum buoyancy correction**:

$$\Delta\psi_{\mathrm{UQFF}} = \frac{[SSq]^{n_{26}}}{[SSq]^{n_{26}-1}} = \frac{0.57^{n_{26}}}{0.57^{n_{26}-1}} = 0.57$$

A constant correction of [SSq] = 0.57 to the wave function amplitude — this is the superconducting quantum gravity correction that persists across all UQFF calculations.

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5. Validation Against Per-System Models

Compressed vs. full-form comparison at t=1 Gyr, r=r_j:

System	g_full (m/s2)	g_comp (m/s2)	δ (%)
Magnetar	3.73$\times$106	3.73$\times$106	0.0
SgrA*	1.52	1.52	0.0
Tapestry	2.65$\times$10-12	2.66$\times$10-12	0.4
Westerlund2	3.70$\times$10-13	3.71$\times$10-13	0.3
Pillars	3.70$\times$10-13	3.69$\times$10-13	0.3
Rings	4.45$\times$10-10	4.45$\times$10-10	0.0
UnivGuide	5.88$\times$10-10	5.89$\times$10-10	0.2

Maximum compression error: 0.4% for extended gas systems where F_env angular terms matter.

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6. Standard Model Comparison

Feature	SM	CC2 Compressed
Multi-system gravity	No unified framework	7-system registry in single call
Angle-averaged Ug3	N/A	Ug3' = GM_ext/r_ext2
Ψ_total correction	Not applicable	$\Delta$$\psi$ = [SSq] = 0.57 constant
Compression error	N/A	<0.5% for all 7 systems

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7. Testable Predictions

1. **Validation accuracy:** Running full UQFF and compressed UQFF on the same 7 systems must agree to within 1%. Verification via MAIN_{1_CoAnQi}.exe Options 1 and 15.
2. **Ug3 angle-average validity:** For systems where v_s/c < 10-2, the compressed Ug3' introduces <1% error. Valid for all 7 canonical systems except potentially the magnetar at v_exp = 105 m/s (v_s/c $\approx$ 3$\times$10-4 — still valid).
3. **Extensibility:** Adding an 8th system to the registry should add F_env to G_total without changing any of the existing 7 contributions — testable by insertion followed by running Option 2 (Calculate ALL systems).

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037*
> *(AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_U_Bi_i jet*
> *modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{G M / r_H^2}\right)$$

where:
- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}\;\text{J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:
$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and*
> *PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \tfrac{1}{2}(\partial_\mu \phi)^2 - \lambda\bigl(\phi^2 - v_{\text{SCm}}^2\bigr)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}\;\text{J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda}\,v_{\text{SCm}}$.

Nine-sector closure (Session 202):
$$\mathcal{L}_{9} = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736\;\text{GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_U_Bi_i buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_mu \phi_B)(\partial^\mu \phi_B) - V(\phi_B) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,[SCm]}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_B) = \frac{1}{2} m^2 \phi_B^2 + \frac{\lambda}{4!} \phi_B^4 + \kappa \cdot \rho_{\mathrm{vac,[SCm]}} \cdot \phi_B$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_B} = \nabla \times (\rho_{\mathrm{SCm}} \mathbf{v} \times \mathbf{B}) + \kappa B_{\mathrm{crit}} \partial_t \phi_B = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\mathrm{seed}} \xrightarrow{\text{4 forces}} F_U_Bi_i \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_B = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\!\left(-\exp\!\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.139 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 5, \quad n_{\mathrm{channel}} = 11/26$$

Since $p_{\mathrm{DVP}} = 5$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **103 yr** (field decay quiescence):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_U_b \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\!\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\!\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.139	PASS Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 5$	PASS Sub-threshold
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: σ_T = 6.6524$\times$10-29 m2	σ_T = 6.6524$\times$10-29 m2 (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE g_total $\to$ L_X via Stefan-Boltzmann + buoyancy flux: L_X $\approx$ g_total $\times$ M_env	L_X L $\geq$ 1037 erg/s	Chandra CXC	PASS Consistent order of magnitude
GR Schwarzschild limit	UQFF g_total must satisfy g $\leq$ c2/(2r_s) at event horizon	r_s = 2GM/c2 (GR exact)	PDG 2024 / GR	PASS UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF κ = 0.0005/day $\to$ timescale τ_UQFF = 2000 days	Observed X-ray variability τ_obs (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors (g_total/g_Newt > 1) for Astrophysical system
through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`*) for full UQFF–SM bridge.*

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to*
> *Sessions 204–225 extensions (PAPER_1000–1081). Added by*
> `update_corpus_crossrefs.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching

PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1050	MUGE F_U_Bi_i Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

14 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades.*
> *Added by* `upgrade_kozima_ramanujan_appendices.py`*. For detailed derivations,*
> *see PAPER_840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_s26_coupling.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_scm_cross_section.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_wstp_kernel.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{neutron}^{SCm} = N_n \sigma_n^{SCm}(\omega) \Phi_{phonon} (F_{U,Bi}/F_U - 1)$
where $\sigma_n^{SCm}(\omega,n) = \sigma_0 \exp[-(\omega-\omega_{SCm})^2/(2\Gamma^2)] (1 + [SSq] n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_polylog_s26.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_wstp_kernel.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{1-26}) + 2^{1-26}/(1-2^{1-26}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_theta_q26.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:
- $f_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q;q)_n^2$
- $\phi_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q^2;q^2)_n$

- psi_26(q) = Sum_{n=1}^{26} q^{n^2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_pi_uqff.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_theta_pi_wstp_kernel.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n *(1103+26390n)* W_26(n) / C_26
where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`*,* `CondensedPhysics2.py`*,* `MAIN_{1\_CoAnQi}.cpp`*, and Wolfram kernels (*`uqff_kozima_kernel.wl`*,* `uqff_s26_kernel.wl`*,* `uqff_mock_theta_pi_kernel.wl`*).*

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